

Human-AI Alignment

Alignment Specification

Alignment Objectives

Task & Assistance Alignment

HelpSteer2 [wang2024helpsteer2]; UltraFeedback [cui2024ultrafeedback]; Scaling Instruction-Finetuned LMs [chung2024flan]; InstructGPT [ouyang2022instructgpt].

Safety Alignment

Lifelong Safety Alignment [wang2025lifelong]; Circuit Breakers [zou2024circuitbreakers]; Rule-Based Rewards [mu2024rulebased]; Safe RLHF [dai2024saferlhf].

Values & Stakeholders

Personalized Alignment

Persona-Judge [zhang2025personajudge]; Personalized Alignment Survey [guan2025personalizedsurvey]; Personalizing RLHF [poddar2024personalizing]; Personalized Language Modeling [li2024personalized].

Pluralistic & Societal Alignment

Steerable Pluralism [adams2025steerable]; Collective Constitutional AI [huang2024collective]; PRISM [kirk2024prism]; Modular Pluralism [feng2024modular].

Uncertainty & Drift

COPR [zhang2025copr]; Lifelong Safety Alignment [wang2025lifelong]; Distributional Preference Learning [siththaranjan2024distributional]; Inverse Reward Design [hadfieldmenell2017ird].

Alignment Supervision

Feedback Source

Human Feedback

MM-RLHF [zhang2025mmrlhf]; RLHF-V [yu2024rlhfv]; HelpSteer2 [wang2024helpsteer2]; Fine-Grained Human Feedback [wu2023finegrained].

AI Feedback

VLFeedback [li2024vlfeedback]; UltraFeedback [cui2024ultrafeedback]; RLAIIF [lee2024rlaif]; Self-Rewarding LMs [yuan2024selfrewarding].

Programmatic & Verifiable Feedback

DeepSeek-R1 [deepseek2025r1]; Math-Shepherd [wang2024mathshepherd]; Rule-Based Rewards [mu2024rulebased]; DeepSeekMath [shao2024deepseekmath].

Feedback & Oversight

Demonstrations & Preferences

HelpSteer2 [wang2024helpsteer2]; UltraFeedback [cui2024ultrafeedback]; RLHF-V [yu2024rlhfv]; Self-Instruct [wang2023selfinstruct].

Critique & Process Feedback

LLaVA-Critic [xiong2025llavacritic]; Let's Verify Step by Step [lightman2024verify]; Math-Shepherd [wang2024mathshepherd]; CRITIC [gou2024critic].

Reliable & Scalable Oversight

Weak-to-Strong [burns2024weakstrong]; AI Control [greenblatt2024control]; LLM Debate [khan2024debate]; Prover-Verifier Games [kirchner2024proververifier].

Alignment Mechanisms

Training-Time Alignment

Reward & Verifier Modeling

BiPRM [zhang2026biprm]; LLaVA-Critic [xiong2025llavacritic]; PRM-Bench [song2025prmbench]; RewardBench [lambert2025rewardbench]; HelpSteer2 [wang2024helpsteer2].

Supervised Alignment

Scaling Instruction-Finetuned LMs [chung2024flan]; SteerLM [dong2023steerlm]; LIMA [zhou2023lima]; Self-Instruct [wang2023selfinstruct].

Preference Optimization

SPPO [wu2025sppo]; SDPO [kong2025sdpo]; MMedPO [zhu2025mmedpo]; KTO [ethayarajh2024kto]; ORPO [hong2024orpo]; SimPO [meng2024simp]; TDPO [zeng2024tdpo]; Direct Nash Optimization [rosset2024dno]; Self-Exploring LMs [zhang2024selm]; DPO [rafailov2023dpo]; IPO/ Ψ PO [azar2023psipo]; HA-DPO [zhao2023hadpo].

Reinforcement Learning

DeepSeek-R1 [deepseek2025r1]; MM-RLHF [zhang2025mmrlhf]; Safe RLHF [dai2024saferlhf]; RLAIIF [lee2024rlaif].

Alignment Distillation

AlignDistil [zhang2025aligndistil]; PAD [gu2025pad]; ADPA [gao2025adpa]; DPKD [li2024dpkd]; PLaD [zhang2024plad].

Inference-Time Alignment

Steering, Search & Refinement

LLaVA-Critic [xiong2025llavacritic]; CRITIC [gou2024critic]; Contrastive Activation Addition [rimsky2024caa]; Self-Refine [madaan2023selfrefine]; Inference-Time Intervention [li2023iti].

Human Control

Corrigibility Transformation [hudson2025corrigibility]; AI Control [greenblatt2024control]; STaR-GATE [andukuri2024stargate]; Learning to Defer [mozanar2020defer].

Alignment Assurance

Evaluation & Robustness

Behavioral & Evaluator Evaluation

PRMBench [song2025prmbench]; RewardBench [lambert2025rewardbench]; SafetyBench [zhang2024safetybench]; HarmBench [mazeika2024harmbench]; HallusionBench [guan2024hallusionbench].

Adversarial & Distribution Robustness

When Context Flips, Safety Breaks [choi2026context]; StrongREJECT [souly2024strongreject]; Circuit Breakers [zou2024circuitbreakers]; HarmBench [mazeika2024harmbench]; XSTest [rottger2024xstest].

Alignment Preservation

Alignment Midtraining [brazilek2026alignment]; Lifelong Safety Alignment [wang2025lifelong]; LARF [li2025larf]; COPR [zhang2025copr]; Fine-Tuning Aligned LMs Compromises Safety [qi2024finetuning].

System Assurance

Mechanistic & Theoretical Evidence

Sustaining AI Safety [mazzu2026sustaining]; Corrigibility Transformation [hudson2025corrigibility]; Contrastive Activation Addition [rimsky2024caa]; Representation Engineering [zou2023repreng]; Inference-Time Intervention [li2023iti].

Monitoring & Auditing

EvalSafetyGap [uluirmak2026evalsafetygap]; RewardBench [lambert2025rewardbench]; AI Control [greenblatt2024control]; Test-Set Contamination [oren2024contamination]; Model-Written Evaluations [perez2023modelwritten].